



Methodological Feedback

Counting South Sudan's cropland

Group [1]

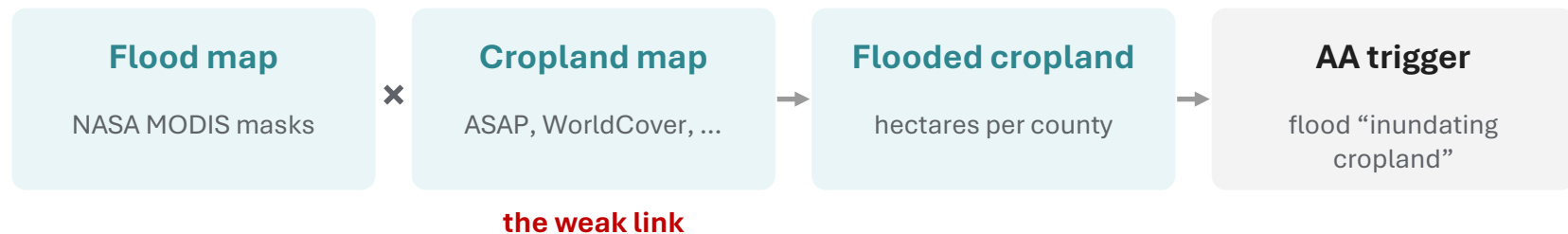
Tutor: [Francisco]





The problem:

a trigger that needs a cropland map



35×

**less flooded cropland than the
FAO/WFP estimate**

South Sudan 2022: flood masks × ASAP ≈
3,700 ha; CFSAM reports 130,000 ha
damaged

0 ha

**flooded cropland ever detected
in Bor South**

every season checked, in an area ZOA
names as flood-prone

***“...might not capture
agricultural land”***

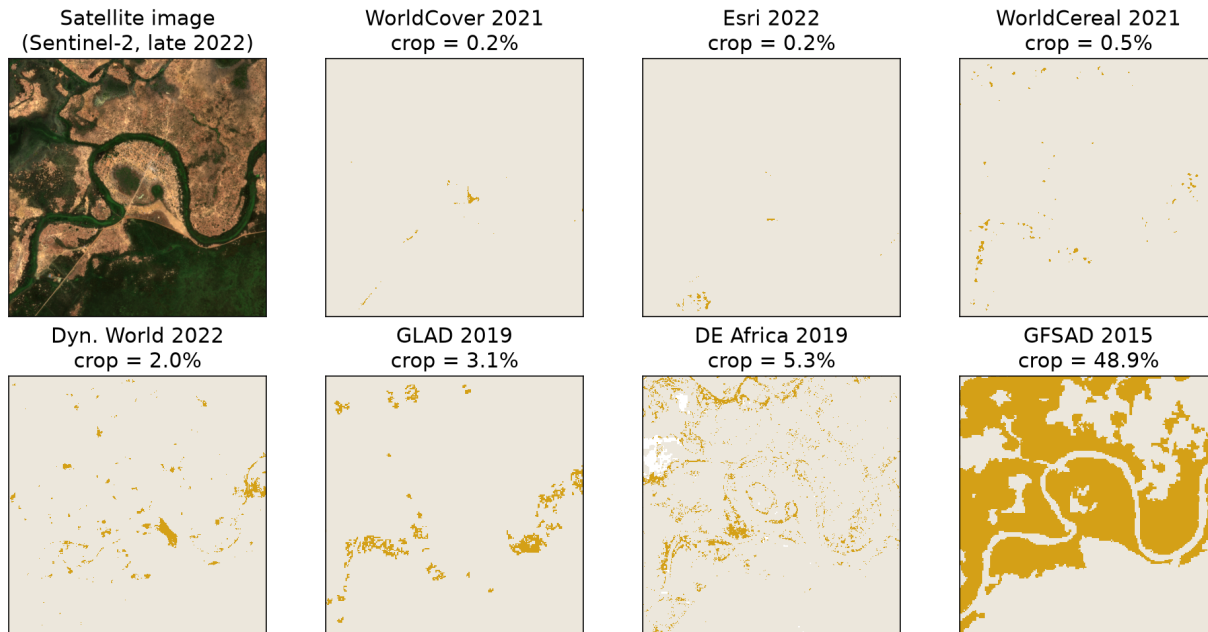
**ZOA about the course cropland
data**

Stakeholder Q&A 2, slide 33



The story:

seven maps, seven answers



0.2%

to

48.9%

crop in the same window

By eye, only GFSAD covers the cleared land, and it is the oldest map (2015).

Not measured yet: that is what our labelled points are for.

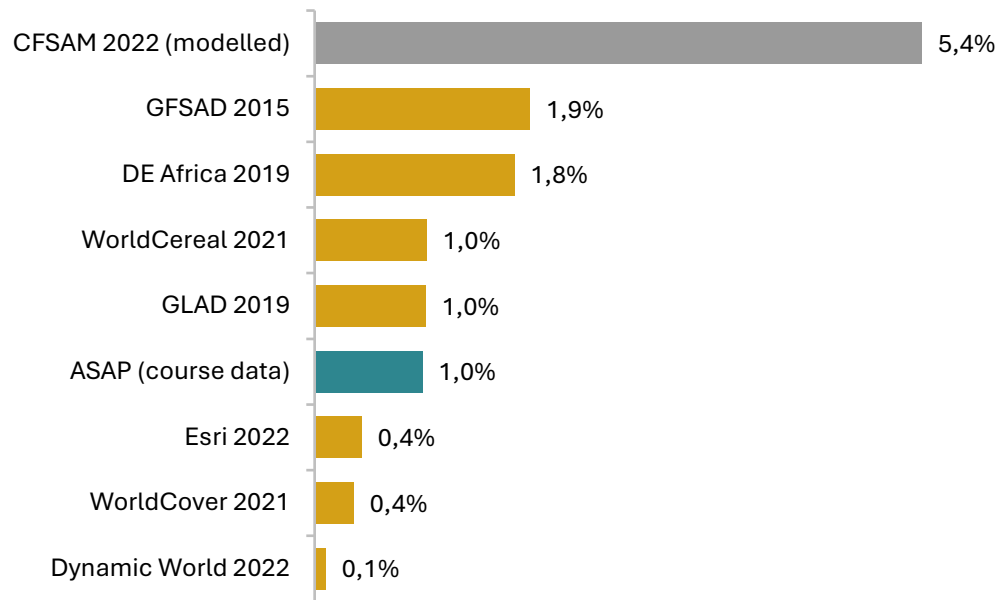
Same 6 × 6 km window near Turalei (Twic county), outside our study counties so the labelling stays blind. Gold = map says crop.



The story:

the maps barely agree, and they are old

Cropland in the 5 Aweil counties (% of land)



1 ha

where the four newest maps (2021–22) all agree on crop, out of 3 million ha. Best overlap of any two maps: $\leq 25\%$.

5×

more crop in the FAO/WFP CFSAM 2022 report than in the typical map. CFSAM is modelled from household statistics, not mapped.

2015–22

map years. GFSAD, DE Africa and GLAD are 2019 or older; ASAP merges older maps. The newest three find the least crop.



Prior work:

how Kerner et al. (2024) audited 11 maps

1

Random sample

Random points per country. In Mali, where crop is rare, stratified by greenness (NDVI) to get enough crop points: 447 points, only 10 were labelled as crop by reviewers.

2

Label by eye, blind to all maps

Monthly 3 m PlanetScope images, Sentinel-2 and Google Earth across the season. Crop = sowing, growth or harvest visible in the greenness time series. ≥ 2 labellers per point.

3

Score every map on the same points

Accuracy, precision, recall and F1 with standard errors. 11 maps, 8 countries, no South Sudan.

Mali: 2.2% of the points are crop

Map	Accuracy	F1
Esri	0.97	0.00
DE Africa	0.96	0.46
WorldCover	0.95	0.43
GLAD	0.96	0.33
ASAP	0.95	0.21
GFSAD	0.91	0.15
Mean of 11 maps	0.92	0.21

Accuracy misleads when crop is rare: Esri scores 0.97 and finds no crop at all. Aweil is like Mali, so we report precision, recall and F1.



Research objective:

what we want to confirm today

Proposed RO

Measure how much cropland ZOA's areas had in 2022, and how accurately public maps, and a readable map of our own, capture it.

1

Estimate of how much cropland?

hectares $\pm$ 95% CI from $\approx$ 450 labelled points, independent of any map

2

How wrong are the public maps?

7 maps scored on held-out points: precision, recall, F1, area

3

Can readable rules do better here?

our 2022 map, scored on the same points

4

Does the choice matter for AA?

2022 flood $\times$ each map: would a trigger have fired?

Scope

Where

5 Aweil counties (Lol river) + Bor South: ZOA's areas

When

2022 season: flooding, and CFSAM figures to compare

Map

whole country, but checked only in ZOA's areas

Out

the flood forecast (documented in prediction/)

Gap

Kerner et al. skipped South Sudan; we found no accuracy check here



Our method:

one labelled sample, two uses



Result A: how much cropland

Stratified estimate on all points (Olofsson et al. 2014). Valid whatever the maps say. Aweil ≈ 330 points $\rightarrow$ about ± 2.5 percentage points: enough to tell 1% from 5%.

Result B: which map to trust

7 public maps + ours on the same 300 test points: precision, recall, F1 and area, with error bars. No map is judged on points it has seen.

Changed from Kerner: we keep points where labellers disagree (report Cohen's kappa), and label from Sentinel-2 and Google Earth because the free Planet imagery (NICFI) ended in 2025.



Our method:

our own map, tested on held-out points

What the labeller sees → what the computer measures

Cue	Feature (from Sentinel-2 / -1)
Bare soil before sowing	low NDVI, bare-soil index Apr–May
Green-up with the rains	NDVI rise Jul–Aug; radar if cloudy
Harvest, grass stays green	NDVI drop Sep–Oct vs 500 m around
Regular plots with edges	texture / edge density at 10 m
Homesteads nearby	distance to buildings
Not crop: trees, water, burns	dry-season NDVI, water, burn scars

Why rules: anyone (ZOA, a reviewer) can read why a pixel is crop; it reruns for any year; radar sees through the July–September clouds. If ours loses, the area estimate still stands.

Three versions, one test

- 1 Hand rules**
baseline: if-then thresholds we set by looking at the 150 points
- 2 Small decision tree**
main candidate: 3–4 levels, still prints as readable rules
- 3 Black box**
benchmark only: random forest or Google's satellite embeddings

Fit on the 150 calibration points → freeze the rules in a dated file → score on the 300 test points, next to the 7 public maps.



Your feedback:

four questions

1

Is this useful to ZOA?

It does not prescribe an AA action. It gives ZOA a checked cropland number and which map to trust, to process and share with communities. Is that a strong enough contribution?

2

How strong is the method?

What would you add? Do you already see a flaw or a gap in the sample, the labels or the test?

3

Is labelling by hand worth it?

$\approx 450 \text{ points} \times 2 \text{ labellers} \approx 22 \text{ hours}$ ($\approx 4.5 \text{ h}$ each). Good use of our time, or is there a shortcut?

4

Is 2022 a useful year?

Many maps stop at 2019 (GLAD, DE Africa). Is a 2022 check useful to ZOA, or would a more recent year matter more?